FYP – OpenPaths

Matthew Pirotta

Supervisor: Dr. Kristian Guillaumier

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L-Università ta' Malta
Faculty of Information & Communication Technology

Contents

List of Acronyms

ABM Agent-Based Modelling.

AT-ABM Active Transportation Agent-Based Model.

BNDP Bicycle Network Design Problem.

C-SAM Connections for Safer Active Mobility.

CC Closest Component.

DNDP Discrete Network Design Problem.

L2C Largest-to-Closest.

L2S Largest-to-Second.

LP Linear Programming.

MILP Mixed Integer Linear Programming.

MPAM Multi-Class Pedestrian Assignment Model.

NHTS National Household Travel Survey.

NSGA-II Non-Dominated Sorting Genetic Algorithm II.

NSO National Statistics Office.

OD Origin–Destination.

R2C Random to Closest.

SPSA Simultaneous Perturbation Stochastic Approximation.

TOPSIS Technique for Order Preference by Similarity to Ideal Solution.

UNDP Urban Network Design Problem.

1 Introduction

Transportation systems influence multiple sectors of society, shaping the environment, public health, and socioeconomic relationships. Decades of car-centric urban development have contributed to rising carbon emissions, escalating road maintenance costs, and the creation of hostile public spaces that discourage active mobility.

A substantial portion of national expenditure is directed toward road construction and maintenance. According to CEIC data [1], Malta spent a median of approximately €21 million annually between 1995 and 2014. Despite these investments, road safety remains a significant issue. National Statistics Office (NSO) [2] reported around 16,000 road accidents in 2023, including 26 incidents involving cyclists, eight of whom sustained grievous injuries. Such figures highlight both the financial and societal costs of car dependency for the country as well as the inherent dangers to cyclists. All of this is further reinforced by the 2030 National Transport Master Plan, which estimates that traffic congestion will cost Malta €770 million in 2025, rising to €917 million annually by 2030 [**?**]. The report attributes these losses to longer journey times, higher vehicle operating costs, and Malta's heavy reliance on private cars, which account for over 84% of all road traffic. It further warns that congestion already costs the country around 3.3% of its GDP, excluding environmental externalities such as emissions and pollution.

Importantly, the same report identifies active mobility, such as cycling and walking, as part of the solution to ease congestion, yet acknowledges that Malta's cycling network remains "fragmented" and "does not penetrate urban areas," with infrastructure that is often "poorly designed" despite recent improvements [**?**]. This highlights a significant gap between the policy vision for sustainable transport and the current state of infrastructure on the ground.

Maltese citizens do not see the country as being apt to cycle in, as reflected in the National Household Travel Survey (NHTS) of 2021 [3], which reported that only 0.7% of all trips were carried out by bicycle. Respondents identified road safety concerns (17%) and poor-quality cycling infrastructure (13%) as the principal barriers to cycling. The Maltese government has shown commitment to improve the local cycling infrastructure in the recent years, with the ongoing project "*Vjal Kulhadd*" launched in 2024 [4], with an investment of 10 million euros [5]. A segregated passageway dedicated to pedestrians and cyclists from Żabbar to Żejtun was completed in March 2025 [6], with the first phase of this project planned to be completed by 2026. Another project is the Connections for Safer Active Mobility (C-SAM) initiative, which is a nationwide cycling and active-mobility network plan for Malta announced in 2022, with an estimated budget of €35 million and a target of 50-60 km of new cycling routes [7]. The Grand Harbour Cycling Network is the first component of C-SAM with a budget of €27.4 million, focusing on the Grand Harbour area (encompassing Msida, Pietà, Blata l-Bajda, Valletta, Marsa and Paola) and is scheduled for completion in phases by 2028 [8].

In parallel, government initiatives are also targeting public transport improvements to reduce reliance on private vehicles. These include the addition of 25 new buses in 2025 [9] and renewed discussions around the proposed Malta Metro system, initially announced in 2021 [10], temporarily shelved, and later revived for consultation in 2025 [11].

Together, these efforts reflect a growing recognition of the need for a balanced and sustainable mobility system, where active and public transport play central roles in shaping Malta's urban future. Transitioning cities toward walkable and cyclable environments can improve public health, reduce emissions, and foster safer and more inclusive urban spaces. To support this transition to a more expansive cycling network, there is a growing need for data-driven tools that assist urban planners in reallocating road space from private cars to active mobility users.

The overarching aim of this project is therefore to develop a computational framework that optimally expands Malta's cycling network, ensuring safe and continuous connections between major destinations while maintaining a functional and connected road system for cars. The current transport context in Malta underscores the urgency of this objective.

2 Literature Review

Designing efficient and safe bicycle networks requires an understanding of how transport systems can be represented, analysed, and optimised using quantitative models. The literature on network design spans a range of approaches that variously emphasise network structure, infrastructure optimisation, and traveller behaviour. Reflecting this diversity, this review first outlines the fundamental concepts common to all modelling frameworks, including graph representation, flow estimation, and network evaluation. Then examining the main optimisation approaches used in bicycle network design: heuristic, linear programming and mixed-integer linear programming, and metaheuristic

2.1 Fundamental Concepts in Road Network Modelling

2.1.1 Graph Representation of Transport Networks

All network design frameworks represent the underlying road infrastructure as a graph, where intersections correspond to nodes and road segments to edges. Across the reviewed studies, network representation varies according to modelling objective. Heuristic or topological studies e.g [12] adopt undirected simple graphs to focus on geometric growth and network efficiency rather than directional flow. In contrast, most optimization-based approaches employ directed graphs, which are the most common representation in transport network modelling because they capture one-way streets and allow for direction level distinction of features, providing a realistic basis for traffic assignment. More advanced frameworks, such as [13], extend this to directed multigraphs that allow lane-level differentiation, where each lane is represented as its own directed edge. Offering finer control over lane-specific attributes such as capacity and flow allocation. This distinction highlights the trade-off between computational simplicity and flow realism in urban network design.

An important extension of this graph representation is the use of multilayer networks, which capture the interactions between different modes of transport. In such models, each layer corresponds to a distinct mode, such as motorised traffic, cycling, or walking, while shared intersections or transfer points act as inter-layer connections. This representation allows planners to account for modal dependencies and conflicts, for example when cycling lanes share space with car traffic or when cyclists use protected corridors separated from the vehicular network. Lonardi et al. [14] explicitly adopt this formulation in their *Cohesive Urban Bicycle Infrastructure Design* framework, modelling the urban transport system as coupled road and protected-cycle layers to evaluate how improvements in one layer affect the other. Even studies that focus solely on the cycling network implicitly rely on this concept, as proposed bicycle links are embedded within an existing multilayered urban fabric that includes vehicle and pedestrian flows.

Such approaches are examples of the broader Urban Network Design Problem (UNDP), and more specifically the Discrete Network Design Problem (DNDP), which is generally considered NP-hard [13]. This complexity arises because network design involves bi-level decision-making. At the upper level, discrete infrastructure choices are made, such as reallocating or upgrading road segments. At the lower level, user flows then adjust to the new network conditions. The interaction between these two levels produces a non-convex and combinatorial search space, making it computationally intractable to find the global optimum for large-scale networks. The way travel demand and flow are represented at this lower level directly affects both the realism of user responses and the computational tractability of the overall optimisation. Accurately and efficiently capturing this interaction is therefore essential for evaluating how proposed network changes influence travel behaviour and system performance.

2.1.2 Flow and Demand Representation

Representing travel demand and flow accurately is a critical element of transport network modelling. In essence, while the graph defines the spatial structure of the transport system, flow representation determines how travellers use that structure, that is, how many trips occur between different locations and how these trips distribute across available routes. There are two main models capturing this relationship at varying levels of detail, Origin–Destination (OD) matrices and trajectory data.

An OD matrix defines the total number of trips between each origin and destination pair within the study area, usually derived from household travel surveys or census statistics. OD data provides a macroscopic view of mobility patterns, making it suitable for equilibrium-based and aggregate network models where flow assignment depends on total demand rather than individual behaviour. For instance, [15, 16] both rely on OD data to simulate pedestrian and cycling flows at a city scale. Another instance, in the Tehran case study, [15] estimated the OD matrix using a combination of household surveys and field observations, such as counting pedestrians at subway stations and bus stops. Trips were classified into home-based and non-home-based categories, further subdivided into work and leisure purposes, allowing the model to distinguish between trip types with different sensitivities to distance and environmental factors. In contrast [12] assume uniform OD demand between all node pairs to isolate the topological effects of network structure.

On the other hand, trajectory data such as GPS traces, or crowdsourced cycling data (e.g., Strava), captures individual-level travel behaviour with high spatial and temporal precision. This data enables the reconstruction of actual routes taken by users, revealing preferences for safety, directness, or comfort that OD matrices cannot capture. For example, [17] employ bike trajectory data to identify observed cyclist movement patterns and use these trajectories to guide the placement of new cycling links, thereby improving the behavioural realism of their model. Trajectory datasets are therefore particularly valuable for agent-based models, where individual route choices and responses to infrastructure are explicitly modelled.

Each data source presents trade-offs. Trajectory data provides detailed behavioural information but often suffers from sampling bias, such as overrepresentation of fitness-oriented or app-using cyclists and privacy limitations, while OD matrices are more systematically collected but lack route-level detail and temporal variation.

Once travel demand and flows are defined through OD matrices or trajectory data, the next step in evaluating bicycle networks is to quantify their structural and functional performance using a range of network metrics or agent based modelling.

2.1.3 Network Evaluation Frameworks

Topological Metrics Network metrics provide quantitative indicators of how effectively a transport network supports mobility, accessibility, and resilience. These metrics are derived from graph theory and in literature are used both to evaluate existing networks and to guide heuristic or optimization models toward more efficient configurations.

Connectedness is among the most fundamental measures, describing whether the network forms a single, navigable system or remains fragmented into multiple components. Metrics such as the size of the largest connected component or the number of disconnected subgraphs are commonly used to assess overall network cohesion. High connectedness indicates that users can reach most destinations without detouring the safe cycle network, an essential property for promoting cycling continuity and accessibility.

Directness measures how efficiently users can travel between two points and is typically defined as the ratio of Euclidean distance to shortest-path distance within the network. Natera et al. [18] take a different approach to directness, calculating it as the ratio of distance travelling with the car compared to the distance travelling with the bicycle. Values closer to one indicate straighter, more direct routes and fewer unnecessary detours, important for both user comfort and competitiveness with motorised travel. Studies such as [12] use directness to evaluate how heuristic growth strategies improve route efficiency as new segments are added.

Centrality metrics capture the relative importance of nodes or links within the network. Betweenness centrality, for example, measures how frequently a link lies on the shortest paths between all node pairs, serving as a proxy for potential flow or load. High-betweenness edges often correspond to corridors that would carry the most traffic if demand were evenly distributed, making this measure a key input in centrality-based heuristics [12]. Other centrality measures, such as closeness (proximity of a node to all others), and degree (number of connections per node), indicate network accessibility and local connectivity density.

Beyond these classical indicators, Szell et al. [12] employ higher-order measures such as global and local efficiency, spatial clustering, anisotropy, and coverage. These capture the balance between resilience, compactness, directional bias, and service reach. For instance, global efficiency measures the overall accessibility of a network by quantifying how efficiently any two nodes can reach each other through the existing connections, while local efficiency evaluates how well a neighbourhood remains connected if a node fails. Spatial clustering and anisotropy help identify concentration patterns and directional preferences, offering insights into the morphological coherence of urban cycling networks. Lastly coverage offers a reasonable buffer where commuters can reach getting out/in the bike network. Unlike purely geometric measures, perceived safety captures how infrastructure design, such as physical separation, traffic volume, or intersection treatment, affects the attractiveness and usability of cycling routes, providing a human-centred dimension to network evaluation.

Together, these metrics provide a quantitative framework for assessing and comparing networks across cities or growth strategies. They are particularly useful for heuristic and optimization approaches that aim to improve network structure without modelling detailed individual behaviour. However, as the next section discusses, agent-based models complement these static indicators by capturing the dynamic, individual-level interactions that give rise to aggregate network flows.

Agent-Based Modelling Agent-Based Modelling (ABM) provides a behaviourally rich framework for analysing active transportation systems. By simulating interactions between individual travellers (agents) and their surrounding urban environment, ABMs operate at the microscopic level, allowing system-wide travel patterns to emerge from the collective behaviour of heterogeneous individuals. Unlike aggregate or topological models, this bottom-up structure explicitly represents decision-making processes such as mode and route choice, enabling researchers to examine how changes in cycling and walking infrastructure influence safety outcomes, travel behaviour, and equity of access. The ability of ABMs to incorporate behavioural adaptation and spatial heterogeneity makes them particularly suited for studying the dynamic relationships between users and the built environment. However, the reliability of their outputs depends on how accurately these models reproduce real-world conditions, which in turn hinges on data quality, calibration methods, and behavioural realism.

Agent-based evaluation methods vary widely in their spatial scale, behavioural detail, and underlying data sources. Aziz et al. [19] developed the Active Transportation Agent-Based Model (AT-ABM) to simulate neighbourhood-scale mode choice between home and work trips. In this model, the agents' baseline probabilities of selecting a transport mode depend on attributes such as income, age, and perceived safety, and are further influenced by social interactions with other agents within the same local environment. The model integrates traffic safety records, capturing pedestrian and cyclist crash frequencies with population and socio-demographic datasets to parameterise agent characteristics. Because the AT-ABM is non-deterministic and lacks analytical gradients, it employs the Simultaneous Perturbation Stochastic Approximation (SPSA) algorithm for heuristic calibration, iteratively perturbing model parameters and evaluating outcome changes to efficiently estimate behavioural sensitivities under stochastic variation.

In contrast, [20] adopt a rule-based ABM to explore how the introduction of separated cycling infrastructure affects driver behaviour and road safety. Using a network of approximately 13,000 road segments, the model represents behavioural adaptation in which drivers gradually learn to anticipate cyclists after repeated encounters, leading to reduced collision risks up to a saturation point. Road types are differentiated by exposure and visibility levels, and potential crash events are simulated at intersections based on whether drivers have recently encountered cyclists. Rather than relying on statistical calibration, parameters such as driver learning rate and perception sensitivity are systematically varied across scenarios to examine emergent safety dynamics. This approach provides a dynamic behavioural perspective absent from traditional static safety models, illustrating how ABMs can capture the temporal evolution of driver awareness and risk perception.

A broader, city-scale application is demonstrated by Jafari et al. [21], who employ an integrated agent-and-activity-based simulation framework to assess changes in travel mode share resulting from proposed cycling infrastructure in Melbourne, Australia. The model makes use of census-based demographic data, OD matrices, and household travel surveys to define behavioural clusters of riders characterised by trip length, safety preference, and socio-economic background. These clusters inform the agent population, allowing simulation of how a proposed cycle corridor would alter travel behaviour across demographic groups. Calibration is achieved deterministically by iteratively adjusting mode-choice constants until simulated mode shares align with observed data. The results show that travel time strongly influences mode choice across all clusters, and that improved cycling infrastructure and perceived safety substantially increase cycling uptake, by approximately 30%, highlighting the value of agent-based approaches for infrastructure impact assessment and policy evaluation.

Collectively, these studies underscore the value of ABMs in capturing behavioural, temporal, and spatial complexity within active transportation systems. By modelling the decision-making processes of individual travellers, ABMs allow researchers to explore how and why infrastructure changes influence user choices, safety outcomes, and modal shifts, moving beyond purely structural or geometric network assessments. Their flexibility makes them particularly powerful for simulating emergent phenomena such as behavioural adaptation, social influence, and mode-shift dynamics under different policy or design scenarios.

Nevertheless, several methodological challenges remain. ABMs are data-intensive, requiring detailed demographic, safety, and travel diary information to parameterise and validate behavioural rules. Their complexity introduces large numbers of parameters, making calibration and validation difficult. Models such as that of Aziz et al. [19] address this through stochastic optimisation techniques like SPSA, while others, such as of Thompson et al. [20], rely on heuristic or manual parameter variation, potentially affecting reproducibility and generalisability. Moreover, many ABMs focus on mode choice and aggregate travel flows rather than explicit route-level movement, which may limit their geometric fidelity in representing real-world network performance. Finally, simplified behavioural assumptions, though necessary for computational feasibility, can constrain realism when modelling diverse urban contexts.

Optimal-Transport-Based Flow Modelling Optimal Transport (OT)–based models provide a continuous, physics-inspired framework for evaluating cyclist movement and network performance. Rather than simulating individual agents or relying solely on topological indicators, these models treat cyclist flows as mass transported along a network under cost-minimising principles. Lonardi et al. [14] formulate cyclist routing as an OT-driven dynamical system in which edge 'capacities' adapt over time to match observed fluxes, allowing the network to converge to weighted shortest-path flows that reflect cyclists' safety preferences, infrastructure coverage, and detour tolerance. Their analysis demonstrates how variations in network completeness affect global metrics such as detour, overlap, and congestion inequality, revealing substantial spatial disparities across Copenhagen's districts. These approaches offer a powerful evaluation tool for understanding how cyclists redistribute themselves under network changes.

These evaluation frameworks provide the means to assess and simulate how effectively a network supports mobility and behavioural adaptation. However, evaluation alone does not determine which infrastructure changes should be implemented. The next challenge is therefore optimisation, identifying from the many possible interventions, the set of new segments or reallocations that most improve overall network performance. The following section reviews the main optimisation approaches proposed for this purpose.

2.2 Optimisation Approaches for Network Design

In the literature, the upper-level optimization problem in bicycle network design is typically addressed through one of three main approaches: heuristic, linear or mixed-integer programming, and metaheuristic methods. Each approach presents distinct advantages, limitations, and suitable use cases, balancing between computational efficiency, behavioural realism, and scalability. With this foundation on how these problems are conceptually structured, the following sections discuss the different modelling approaches in greater detail.

2.2.1 Heuristic Approaches

The simplest of the methods used in bicycle network planning are heuristic approaches. These are considered "simple" because they rely on rule-based or greedy decision-making guided by intuitive network metrics such as connectedness, directness, or betweenness centrality. Heuristic models iteratively add or remove edges in the bicycle network to improve a selected topological metric, without formulating the problem as a formal bi-level optimization task. In doing so, they make several simplifying assumptions, most notably, that traffic demand is uniformly distributed across the network and that travellers do not adjust their routes in response to infrastructure changes (i.e., no user equilibrium is modelled). As a result, these methods capture the geometric and structural characteristics of a network but neglect behavioural feedback and flow dynamics.

For instance, Szell et al. [12] adopt a centrality-based heuristic, where new links are added according to graph-theoretic importance measures, such as betweenness or closeness centrality. This approach prioritizes edges that are topologically critical or spatially central within the underlying street network, allowing the simulated bicycle network to grow along corridors that would most efficiently shorten travel distances.

In contrast, Natera et al. [18] propose component-based heuristics designed to reconnect fragmented pieces of existing infrastructure. Their Largest-to-Second (L2S)strategy links the two largest disconnected components to quickly consolidate the network, while the Largest-to-Closest (L2C) strategy connects the largest component to its nearest neighbour, favouring shorter and more spatially efficient connections. Both approaches share the same principle of incremental, topology-driven expansion, but differ in how they define the "importance" of the next edge to add: L2S biases towards overall connectedness, whereas L2C favours directness by preferring shorter spatial gaps. The authors also use simpler baselines for comparison, using Random to Closest (R2C), representing uncoordinated development, and Closest Component (CC), which connects the nearest two subgraphs, producing a minium spanning tree that minimises investment but is not resilient to any interruptions. Overall, L2C performed best, with L2S also strong but slightly less effective, both outperforming the baselines. The study shows that targeted investments into key missing links can rapidly consolidate fragmented bicycle networks, though cars still achieve greater directness, with routes averaging about 33% shorter than those of bicycles.

A third paradigm is introduced by Steinacker et al. [22] applies percolation-based heuristics to model network growth and robustness over time. In a forward percolation approach, the network gradually expands by adding segments with the highest marginal contribution to global accessibility or welfare. Conversely, the inverse percolation approach begins from a fully built network and iteratively removes the least critical segments, recalculating connectivity and demand after each step. An additional advantage of the inverse percolation method is that it inherently maintains full network connectivity throughout the process, as segments are only removed if their deletion does not fragment the network. This ensures that all resulting configurations remain fully functional while progressively identifying the most critical corridors. These methods provide a transparent way to approximate how incremental investments or disinvestments affect overall network integrity.

Heuristic models consistently outperform baseline random growth strategies. Centrality-based growth yields higher connectedness, directness, and resilience than both random and minimum spanning tree baselines, producing networks that more closely resemble those of well-developed cycling cities such as Copenhagen [12]. Component-based heuristics (L2S and L2C) outperform random or closest-component baselines by achieving faster consolidation and greater overall cohesion [22]. These findings demonstrate that, even under uniform demand assumptions, simple topology-driven heuristics can reproduce high-quality, cohesive bicycle networks far more efficiently than unguided or piecemeal local planning.

While all heuristic families improve structural aspects of the network, they share fundamental simplifying assumptions. All assume uniform travel demand across the city and neglect behavioural factors such as route choice, mode preference, or induced demand. Centrality-based growth interprets betweenness as a proxy for flow under uniform conditions, component-based heuristics connect infrastructure purely by geometry, and percolation-based methods evaluate robustness statically without modelling user adaptation. Consequently, heuristic approaches provide a computationally efficient and transparent means of exploring large-scale bicycle networks and identifying missing or critical links, but their simplicity comes at the cost of behavioural realism, dynamic feedback, and global optimality.

2.2.2 Mathematical-Optimisation Approaches

Mathematical-optimisation approaches including Linear Programming (LP), Mixed Integer Linear Programming (MILP), and related variants represent the most analytically rigorous class of methods for solving the Bicycle Network Design Problem (BNDP). These models treat network design as a constrained optimisation problem in which infrastructure decisions are chosen to minimise or maximise explicit objective functions (e.g., travel time, continuity, or welfare) while respecting physical, spatial, and budgetary constraints. Unlike heuristic approaches, which rely on structural rules or centrality metrics, optimisation-based methods search a well-defined feasible space and provide transparent trade-offs between competing goals such as cycling accessibility, car-network performance, construction costs, and equity considerations.

LP formulations optimise continuous variables, such as allocating a street's lane capacity between cyclists and cars. MILP extends this framework by introducing binary or integer decision variables that capture discrete choices, including whether a protected bike lane is installed on a link or whether an upgrade is built along a cyclist trajectory. This increases modelling realism but also substantially raises computational complexity. Large-scale MILPs are NP-hard, often requiring relaxation, decomposition, or iterative rounding techniques to remain tractable [13]. Furthermore, most LP/MILP models rely on static route choice and fixed OD matrices, assumptions that simplify computation but limit behavioural realism compared to agent-based or dynamic assignment models.

Several studies emphasise the computational intractability of bicycle network design. Wiedemann et al. [13] note that even subtasks of their model—such as optimising lane direction via edge flipping to maintain connectivity—are already NP-hard, implying that the full lane-allocation problem is at least equally difficult. Liu et al.'s [17] trajectory-based MILP further inherits NP-hardness through its binary knapsack link-selection variables, and Rastbin et al. [15] explicitly classify their bi-level pedestrian–cycle optimisation framework as NP-hard. Collectively, these works confirm that urban bicycle network design lies firmly within the NP-hard class, motivating the use of relaxations, heuristics, and metaheuristic algorithms to obtain tractable solutions for real-world networks.

Linear Programming Wiedemann et al. [13] propose a multi-criteria linear programming model that allocates limited street capacity between cars and cyclists. The model minimises the combined travel time of both modes, weighted by a tunable parameter that expresses the policy preference between car and bicycle performance. To reduce computational cost, the authors apply a linear relaxation of the integer lane-allocation problem and iteratively fix the highest-capacity edges after each run, progressively reducing the search space. The final model is able to handle 830-edge networks on a single core up to 10,000 edge networks using parallel processing. The approach produced a 35 % improvement in cycling performance with less than a 20 % increase in car travel time, outperforming heuristic baselines.However, the model tends to allocate one lane of each two-lane road to cyclists, neglecting detailed congestion dynamics for cars. Finally, the result took about 16 hours to reach.

Mixed-Integer Linear Programming The trajectory-based BL-AC and BL-GU models [17] formulate bicycle network design as a true MILP that integrates observed cyclist trajectories into the decision process. Each candidate road segment is associated with a binary construction variable, allowing the model to explicitly represent whether a proper bike lane is installed. BL-AC (Adjacency Continuity) maximises the number of pairwise-adjacent constructed segments, producing geographically dispersed upgrades with broad coverage. In contrast, BL-GU (Generalised Utility) builds longer, more cohesive corridors by maximising a supermodular utility function tied to the size of continuous constructed segments. The models incorporate travel-time disutility, cyclist preferences, and—when extended with a route-choice layer—Multinomial Logit (MNL) probabilities that account for how users may alter their routes in response to infrastructure changes. Empirically, BL-AC yields higher coverage but shorter continuous paths, whereas BL-GU produces substantially larger contiguous corridors (up to 49 connected segments compared to 21 in BL-AC). Including route choice causes infrastructure to geographically spread out rather than forming a single dominant spine, underscoring the importance of behavioural feedback in MILP-based planning. While computationally more demanding than LP relaxations, these models demonstrate how discrete link-level decisions and behavioural realism can be embedded into optimisation-based network design.

Related Mathematical-Optimisation Approaches Direct Optimisation Steinacker et al. [22] extend optimisation-based bicycle network design by introducing a temporal decision-making framework that determines not only where, but also when, segments of Copenhagen's cycle-superhighway network should be built. Unlike a single global MILP that optimises all years simultaneously, their method solves a sequence of independent annual optimisation problems, each selecting the next set of links that maximises time-discounted welfare (Net Present Value) given the infrastructure already constructed. This "direct optimisation" approach therefore yields year-by-year optimal decisions, but does not guarantee global optimality across the entire planning horizon. Instead, it provides a computationally tractable way to incorporate phasing, discounting, and evolving network conditions into the planning process. Within the broader landscape of LP/MILP-style models, Steinacker et al. demonstrate how temporal, economic, and multimodal considerations can be embedded into optimisation frameworks without constructing a fully time-expanded MILP, which would be computationally prohibitive for decades-long horizons.

While Steinacker et al. [22] report that their direct-optimisation strategy yields marginally higher welfare than the heuristic benchmark, the principal differences arise in the temporal sequencing of construction rather than the overall extent of the resulting network. Their uncertainty analysis further shows that demand variation has minimal influence on welfare outcomes, whereas cost uncertainty substantially affects optimisation-driven plans.

Synthesis While these models guarantee mathematical optimality and offer interpretable policy trade-offs, their precision comes at a computational cost. Solving large-scale LPs\MILPs with spatial or behavioural constraints remains NP-hard, often requiring simplifications such as relaxations (temporarily allowing integer variables to take continuous values), decomposition (solving smaller subproblems), or iterative rounding (progressively fixing variables) to maintain tractability [13]. These simplifications improve computational efficiency but sacrifice global optimality, resulting in solutions that approximate the best network configuration rather than guaranteeing it. Moreover, most LP\MILP based formulations generally rely on simplified behavioural assumptions, for example, fixed OD demand, where the total number of trips between origins and destinations remains constant, and static route choice, where users follow a single optimal path that does not adjust to changing network conditions. These assumptions make MILP frameworks computationally efficient but limit their ability to capture behavioural feedbacks such as induced demand, route adaptation, or mode shifts, which are better represented in agent-based models.

Together, these optimisation-based methods demonstrate how linear, mixed-integer, temporal, and dynamical extensions enhance the policy relevance of bicycle network planning. LP and MILP formulations offer globally coordinated and interpretable solutions but rely on simplified behavioural models; temporal direct-optimisation frameworks incorporate economic phasing and uncertainty; Despite their limitations, optimisation-based approaches deliver transparent, reproducible, and competitive alternatives to heuristic methods, and continue to form the backbone of analytically rigorous bicycle network design research.

To address the computational challenges, research has explored metaheuristic optimization that can handle multi-objective problems more flexibly.

2.2.3 Metaheuristic Approaches

Metaheuristic approaches are a less explored class of methods that aims to balance realism and computational feasibility, modelling the real world more accurately than simple heuristics while avoiding the exhaustive search required by exact MILP formulations. To achieve this, [15, 16] employ the non-dominated Sorting Non-Dominated Sorting Genetic Algorithm II (NSGA-II) to solve multi-objective optimization problems that simultaneously consider conflicting objectives for non-motorist and car users. NSGA-II is an evolutionary algorithm that iteratively evolves a population of candidate solutions, ranking them based on their Pareto dominance i.e. how well they perform relative to others across multiple objectives without being strictly better or worse in all of them. The outcome is a Pareto front, a set of optimal trade-off solutions where improving one objective (e.g., cyclist safety or travel time) would necessarily worsen another (e.g., car travel efficiency). Within this front, crowding distance is used to preserve diversity among non-dominated solutions, preventing premature convergence to a single region of the search space. After convergence, the final non-dominated set is further refined using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method to rank and select the most balanced designs. This approach enables planners to explore a spectrum of balanced network configurations rather than a single, rigid optimum.

In Rastbin et al. [15] when evaluating the model on the large-scale case study on the historical-Cultural district of Tehran, Iran (approximately 2,000 hectares, 55 nodes, and 98 edges), the proposed NSGA-II bi-level model demonstrated strong computational efficiency and scalability. The lower-level Multi-Class Pedestrian Assignment Model (MPAM) was solved with an accuracy threshold of 10^3, ensuring convergence while maintaining tractability for the large network. The optimization process required approximately 8,100 MPAM evaluations, compared to over 1,048,000 evaluations that would be needed under full enumeration, representing a significant reduction in computational effort. The test network included roughly 1,800 OD pairs representing both work and leisure trips. The resulting Pareto front provided planners with a spectrum of trade-off solutions balancing travel cost, environmental quality, and construction cost, confirming that the NSGA-II approach can efficiently solve complex, behaviourally informed network design problems at a practical scale.

3 Different features, car safety, grade

- Route Choice - induced demand - OD flows, and time varying flows (nuh uh) - grade - bi level

4 Comparing Results Across the Literature

Direct comparison of quantitative results across bicycle-network design studies is inherently difficult because the papers differ in four fundamental dimensions:

- **Problem formulation**. Studies do not optimise the same thing. Some grow networks from scratch [12, 18, 22], others reallocate existing road space [13, 17], and some mix capacity reallocation with new links [22]. Scratch-based approaches measure idealised structural improvements, whereas reallocation models reflect real-world constraints; their outcomes therefore quantify different phenomena.

- **Metric definitions**. Even when papers use the same terminology, underlying definitions differ. For example, "directness" in Szell et al.[12] is the Euclidean–shortest-path ratio, whereas in Natera Orozco et al.[18] it compares bicycle paths to car paths. Although both are labelled "directness", the former measures geometric efficiency while the latter captures bicycle–car competitiveness.
- **Network size and morphology**. Case studies range from small districts with ∼100 edges [15] to large urban networks with over 14,000 edges [14]. Network size and street morphology strongly influence centrality, connectedness, and efficiency metrics, making their values non-comparable across cities.
- **Flow and demand assumptions**. Studies employ fundamentally different demand models: uniform OD matrices [12, 18], survey- or observation-based OD matrices [15, 21], or real cyclist trajectories [17]. Since network performance depends on routing behaviour and demand distribution, these choices significantly alter metric outcomes.

Taken together, these methodological differences mean that numerical performance measures across studies do not represent the same underlying constructs, and cross-paper numerical comparisons should therefore be interpreted qualitatively rather than quantitatively.

Comparison of Results Across Approaches The reviewed studies report a wide range of outcomes depending on the modelling approach adopted, with clear differences emerging between heuristic, MILP/LP and metaheuristic. Heuristic approaches such as Szell et al.'s growing-network model[12] show that simple rules like betweenness or closeness growth can rapidly consolidate cycling networks: the Largest Connected Component expands quickly, global efficiency rises to around 0.82 (approximately three times that of existing networks), and synthetic networks significantly outperform current real-world layouts. However, these benefits occur without behavioural realism, and car-network effects remain minimal except for a modest ∼5% increase in clustering under the fietsstraat assumption. A similar pattern appears in [18], where small, targeted missing-link additions increase Bogotá's cycling connectedness from 15% to 56% and improve bicycle-to-car directness from 6% to 48%, though bicycles remain roughly one-third less direct than vehicles.

In contrast, linear and mixed-integer programming models provide more coordinated and globally optimised interventions. Wiedemann et al.'s LP framework [13] demonstrates that reallocating lanes can decrease perceived bicycle travel times by over 35% while limiting car travel-time increases to under 20%. Integer relaxation introduces only a ∼5% optimality loss and enables computation on networks with up to ∼10,000 edges when parallelised. Structurally, when rebuilding the whole city from scratch the method converts roughly 60% of streets into one-way operations and reassigns about one-third of motorised lanes to cycling. Other MILP studies, such as Steinacker et al.'s long-term cycle-superhighway planning [22], find that both direct optimisation and backward percolation yield similar final network sizes (1300–1900 km), but direct optimisation achieves 8–10% higher welfare (NPV) by year 30. This gain comes with higher sensitivity to parameter uncertainty, whereas backward percolation is more robust but yields lower welfare early on. Trajectory-based MILPs (BL-AC and BL-GU) [17] further reveal trade-offs between coverage and continuity: BL-AC installs more segments but results in weaker continuous corridors, while BL-GU produces up to 49-segment high-continuity paths but covers fewer streets. Incorporating route choice spreads infrastructure geographically rather than producing a single dominant backbone, illustrating the behavioural impacts of infrastructure allocation.

The metaheuristic algorithm evaluated in Rastbin et al. [15] assesses its performance using a welfare-gain metric, defined as the improvement of each objective relative to the current network, normalised by the best value that objective could achieve if optimised in isolation. Using this measure, the model achieves an 84.24% welfare gain for F1 (total travel cost), indicating a substantial reduction in pedestrian travel effort. The environmental-quality objectives (F4)—including security, safety, walkability, sociability, sense of richness, and permeability—show similarly strong gains, ranging from 78% to 91%, demonstrating consistent enhancements across all qualitative dimensions of the pedestrian environment. Taken together, the average welfare gain of roughly 84% highlights the effectiveness of multi-objective optimisation in producing balanced improvements across conflicting criteria. Although the study focuses on pedestrians, the underlying insight—that carefully structured multi-objective methods can yield large, well-distributed network improvements—is directly relevant to cyclist-oriented infrastructure design.

4.1 Conclusion

The reviewed studies demonstrate that bicycle network design can be approached through a range of methodological frameworks. Together, these methodological perspectives reveal a landscape of complementary tools rather than a single dominant paradigm. However, a persistent challenge lies in integrating behavioural realism and computational scalability within a unified optimisation–evaluation framework. Addressing this gap motivates the present work, which aims to develop a modelling tool that combines scalable network optimisation with behaviourally informed evaluation to support evidence-based planning of cycling infrastructure in Malta.

Biggest question I'm trying to answer: Can maltese roads be reallocated to cyclists

5 Methodology

5.1 Network Representation

A critical step in the modelling pipeline is selecting an appropriate graph representation for the transport network. As outlined in section 2, the literature employs three main structures: undirected graphs, directed graphs, and directed multigraphs. The objective of this project is to propose a practical and implementable cycling network for Malta that improves bicycle connectivity while minimising disruption to the existing car network. For this reason, the transport network was modelled as a directed graph, which aligns with the majority of optimisation-based studies and more accurately reflects real-world travel patterns by capturing one-way streets and direction-specific attributes essential for OD-based flow assignment.

An undirected representation was deemed unsuitable, as it cannot encode directional restrictions or asymmetries in travel behaviour, both of which are common in Malta's dense urban street layouts. Directed multigraphs were also considered but ultimately not adopted. While they allow lane-level modelling of capacity and flow, this level of granularity exceeds the scope of the present study. Additionally, modelling lane-by-lane flow would introduce substantial computational overhead as the number of edges in the graph substantially increase, especially on the scale considered of all of Malta. A directed graph therefore provides the appropriate balance between behavioural realism, data availability, and computational tractability.

5.2 Feasibility Constraints for Lane Reallocation

Another critical step is selecting the problem formulation most appropriate for the Maltese context. As discussed in the literature, bicycle network design can be approached in three main ways: (i) creating new links between nodes (i.e., building new roads), (ii) reallocating space within the existing road network, or (iii) a hybrid of both strategies. For this project, the reallocation approach was chosen. This decision is motivated by Malta's severe spatial constraints. As a densely built island with limited remaining open space, expanding the road network by constructing new corridors is neither practical nor desirable, particularly given the importance of preserving the remaining green areas and avoiding further overdevelopment. A reallocation formulation therefore aligns with environmental, spatial, and policy considerations while constraining the search space to the existing network. Computationally, this avoids the need to generate and evaluate new edges, making the optimisation problem more tractable and better grounded in realistic planning constraints.

However, restricting interventions to existing links also reduces design flexibility. Without the ability to introduce new connections, the model cannot overcome structural bottlenecks inherent in the current street layout. This limitation is particularly evident in areas of Malta where residential grids consist entirely of single-lane, one-way streets (**??**, circle A). In such configurations, it is mathematically impossible to reallocate lanes to cycling without fragmenting the car network, because the street geometry provides no redundant capacity or bidirectional alternatives. Maintaining strong connectivity under reallocation requires corridors with at least two lanes or paired opposing one-way streets.

This issue is not unique to the present study: even advanced optimisation frameworks struggle under the same constraint. Wiedemann et al.'s LP formulation [13] explicitly notes that their algorithm is predisposed to reassign *one lane of any double-lane road* to cycling while preserving feasibility for cars. Their method guarantees strong connectivity only because the capacity constraint is bounded by the number of lanes; when streets have a single lane, the model cannot reallocate space without violating car reachability. In other words, their formulation also relies on the presence of multi-lane corridors and would fail in a network dominated by one-lane streets—precisely the condition found across many Maltese localities.

Similar challenges arise in regions such as circles B and C in **??**. Although a few bidirectional edges exist, many critical corridors still contain only one outgoing lane and function as articulation edges; reallocating them would break reachability and disconnect parts of the car network.

While some streets could technically be reallocated for cycling, constructing a continuous and fully segregated protected bicycle network is often infeasible under these structural constraints. To address this, the project adopts a *fietsstraat* (bicycle street) approach, similar to [12], for segments where lane reallocation would compromise car connectivity. In fietsstraten, cyclists and cars share the roadway, but cyclists receive priority and speed limits are reduced, improving safety while preserving the strong connectivity of the car network.

An additional strategy that was considered was to parametrise the model to increase the effective road capacity of single-lane streets by reallocating space currently used for on-street parking. In principle, removing parking could free sufficient road width to accommodate an additional traffic lane or a protected cycling facility. However, implementing such an approach was not feasible because the necessary data are not publicly available. OSM does not consistently provide road-width measurements or detailed information on the presence and extent of on-street parking along each segment. Without reliable width or parking data, any capacity expansion would require speculative assumptions, undermining the validity of the optimisation results. For this reason, the model focuses solely on reallocation of existing lane capacity rather than inferring latent space from parking removal.

5.3 Preprocessing

5.3.1 Network Extraction & Information Representation

Network Extraction The underlying street network of Malta was obtained through the publicly available OpenStreetMap (OSM) API. When loading the network, the parameter `simplify=True` was used. This option streamlines the graph by collapsing sequences of non-intersection nodes into single edges while preserving the full geometry of the original road segment as an edge attribute. This substantially improves performance, as the number of nodes and edges is greatly reduced, lowering the computational cost of operations performed on the graph.

However, if a simplified edge corresponds to multiple underlying OSM ways, their distinct attributes are retained as lists. While this aggregation is useful for reducing graph complexity, it introduces irregularities in attributes that normally should take a single value. For example, two merged edges may carry highway tags "residential" and "primary", resulting in a merged edge with `highway=["residential","primary"]`. A single classification is required for downstream processing, particularly for safety-related evaluations. To resolve these conflicts, the least safe (i.e., highest-risk) tag was selected as the final attribute. This conservative choice reflects a cyclist-centred perspective: starting from the worst-case classification ensures that subsequent infrastructure allocation prioritises safety.

The parameter `retain_all=True` was also enabled to ensure that the entire OSM network is returned, including any disconnected subgraphs. This is important because the protected cycling infrastructure in Malta is highly fragmented, and discarding disconnected components at this stage would remove most of the existing bicycle network.

Modal Subnetwork & Master Graph Construction Two overlapping but distinct subnetworks were then extracted from the OSM data: the bikeable network, denoted as G_b, and the drivable network, denoted as G_c. Each edge in these subnetworks was annotated with Boolean attributes `bike_allowed` and `car_allowed`, indicating whether the edge is traversable by the corresponding mode. These attributes allow consistent modal tracking throughout the processing pipeline.

The two subnetworks were subsequently merged into a single composite master graph, G, which serves as the unified source of truth for all later stages of analysis. This integration is necessary because several edges are shared between the car and bicycle networks, and maintaining a single combined graph avoids inconsistencies when reallocating lanes or updating network attributes. After composition, only the largest weakly connected component of $G$was retained, as the project focuses solely on lane reallocation rather than constructing new roads. Consequently, any node unreachable in the composed graph would remain unreachable in all future network configurations.

However, `OSMnx` provides and expects multi digraphs, typically these 'parallel' connections in the same direction are roads that have a physical barrier between them. These cases are rare ($<1\%$ in the case of Malta) and the lost of just a bit of information is worth for a more standardised approach. This is primarily done to have a standardised representation of the network. The multi digraph is converted to a digraph while retaining as much information as possible. In the case where two edges in the same direction between nodes occur these must be aggregated into a single edge. For numerical attributes such as length and grade an average value is taken to best

NOTE that all roads on the same edge should be the same type. I.E you cant have a freistatt and 'normal' car road on the same edge. And it doest make sense to have different speed limits on the same edge.

5.3.2 Attribute Enrichment

Graph Projection Before further preprocessing, the network was projected from geographic coordinates (latitude–longitude) to a planar coordinate reference system (CRS). Projection is essential because shortest paths and distances cannot be computed accurately using unprojected spherical coordinates, where units are expressed in degrees rather than metres. OSMnx automatically selects a suitable projected CRS for the study area based on its geographic location, choosing the local UTM zone that minimises distortion. This ensures that all spatial operations—such as distance calculations and edge lengths are carried out with consistent metric accuracy across the entire Maltese network.

Elevation and Grade Once the master graph was obtained, it was preprocessed and standardised to ensure that all required attributes were present. Elevation data for each node was retrieved using the Open Topo Data API, which provides values from the European Digital Elevation Model (EU-DEM) at a 25 m resolution. This was the finest publicly available elevation dataset for Malta, and its resolution is sufficient for urban-scale cycling analysis given the island's relatively gentle topography and absence of steep mountainous terrain.

However, not all nodes returned elevation values, so missing entries were imputed by averaging the elevations of their neighbouring nodes. This interpolation was performed iteratively because some nodes had neighbours that also lacked elevation data and required values to be propagated, until no further updates were possible, effectively producing a local linear interpolation. This neighbourhood-based approach was preferred over using a global mean, as elevation varies spatially and preserving local gradients is essential for computing realistic slope values. With elevation assigned to all nodes, the grade (steepness) of each edge was then calculated using the standard rise-over-run formulation. During this process, a small number of edges were found to have implausible gradient values, with absolute grades exceeding 100% (i.e., slopes steeper than $45°$), which are infeasible for road infrastructure and likely resulted from residual elevation artefacts. To ensure physical realism and prevent these outliers from skewing downstream analyses, grades were clamped to the range $\pm 20\%$ (approximately $11°$). This upper bound remains challenging for cyclists but represents a more realistic maximum slope found within Malta's road network.

Road-Class Consolidation Another preprocessing step involved merging semantically equivalent road classifications. For example, OSM distinguishes between "motorway" and "motorway_link", where the latter refers to short connector segments between major roads. For the purposes of this study, such distinctions are not meaningful, as both road types present a similar level of exposure and danger to cyclists. These categories were therefore consolidated to ensure consistent treatment during network analysis and lane reallocation.

Safety Classification Another preprocessing step involved classifying the safety level of each edge. Road segments where cyclists travel alongside motor traffic were labelled as cautious, reflecting the elevated exposure to vehicles. Primary roads and trunk roads were classified as dangerous, as these typically feature high traffic volumes, multiple lanes, and higher speeds. Additionally tunnels are also considered unsafe due to reduced visibility. Conversely, edges that are fully dedicated to bicycle use—such as segregated cycle paths or shared-use paths without vehicular access—were labelled as safe.

Reallocatability Finally, each edge was assigned a `reallocatable` Boolean attribute. An edge was marked as non-reallocatable if it did not belong to the car network or if bicycles were not permitted on that segment (i.e., it does not appear in the bikeable network and is classified as dangerous). All remaining edges were considered reallocatable.

5.3.3 Summary

In summary, after the preprocessing pipeline the final network is represented as a directed graph whose edges contain all attributes required for subsequent optimisation. Each edge stores information obtained from OSMnx and Open Topo Data, including length (in metres), the number of car and bicycle lanes, and node elevations. From these, additional attributes are derived, such as edge grade, safety classification, and whether the segment is eligible for reallocation by the optimisation algorithm.

5.4 Optimisation Algorithms

To evaluate which optimisation strategy produces the most effective cycling network for Malta, three classes of segment-selection algorithms were implemented: heuristic, mathematical, and metaheuristic approaches.

Objective 1 - Heuristic The heuristic methods follow directly from the techniques reviewed in section 2.2.1. Centrality-based heuristics were implemented using edge betweenness and closeness centrality, prioritising segments that are topologically critical or spatially central [12]. In addition, the component-based heuristics of [18] were replicated, including the L2S, L2C, R2C, and CC strategies. These heuristics aim to consolidate fragmented cycling infrastructure by reconnecting components in a manner that balances cohesion, spatial efficiency, and investment cost. Additionally, the c

Although some studies [12, 13, 22] explore backward (bike→car) strategies to avoid path-dependence and analyse growth sequences, these methods are primarily valuable for heuristic benchmarking rather than practical planning. In this project, the primary goal is to design a forward-looking cycling network for Malta, starting from the current car-dominated layout. Since LP/MILP approaches already provide globally coordinated allocations without relying on sequential growth, implementing backward strategies for these optimisation models would add substantial complexity with little analytical benefit. Similarly Wiedemann et al. [13] only implemented backward heuristic strategies for better comparability of models.

Objective 2 - Mathematical Approaches The second objective is to formulate the network design task using the more rigorous mathematical optimisation approaches discussed in section 2.2.2. This involves constructing a linear or mixed-integer linear programming model that computes a globally optimal solution subject to the constraints of the Maltese road network. Compared to the heuristic strategies, the mathematical formulation incorporates additional detail, including edge grade, mode-specific perceived travel times, and origin–destination travel demand between localities. The OD matrix used in this model will be derived from Saliba et al. [23], who inferred travel flows for Malta using anonymised Call Detail Record (CDR) data from GO.

Objective 3 - Metaheuristic The third objective is to implement a metaheuristic optimisation framework capable of exploring the solution space more flexibly than exact mathematical programming. Building on the approaches discussed in section 2.2.3, a multi-objective evolutionary algorithm, NSGA-II will be applied to balance competing criteria—including cycling accessibility, car travel performance, and network safety—without requiring the strict linearity or relaxations imposed by LP/MILP models. The metaheuristic formulation evaluates candidate network configurations using the same enriched attribute set (grade, perceived travel times, and OD-based flows) but searches the design space stochastically through mechanisms such as mutation, crossover, and non-dominated sorting. This allows the identification of a diverse Pareto-optimal set of solutions, enabling planners to assess trade-offs across different policy priorities rather than converging to a single prescriptive optimum.

5.5 Evaluation

During each iteration when a new segment is suggested by the model, the performance of the new street layout is measured through topological metrics on the LCC of the safe bikeable network. The LCC is chosen by the total length in the edges and not the number of nodes, since edges can be of different length, and length is much more meanigful than the number of intersections. The connectedness is evaluated through the number of components, and the lcc length. For centrality the the mean edge betweenness centrality and closeness centrality are taken. Lastly directness is also calculated.

5.6 Evaluation

At each iteration, when the optimisation model proposes a new segment for inclusion in the cycling network, the resulting street layout is evaluated using a set of topological metrics applied to the LCC of the safe bicycle subnetwork. The LCC is defined by total edge length rather than node count, as length better reflects meaningful network extent, particularly when intersection density varies across areas.

Connectedness is assessed through two measures: the number of connected components in the safe network and the total length of its LCC. These metrics capture whether the proposed interventions reduce fragmentation and extend continuous, navigable cycling corridors.

Centrality is evaluated using the mean edge betweenness centrality and mean edge closeness centrality across the LCC. Betweenness captures the structural importance of edges for shortest-path routing, while closeness reflects overall accessibility within the component. Higher values indicate networks that distribute movement efficiently and minimise detours.

Directness is computed as the ratio between Euclidean distance and shortest-path distance along the safe cycling network, as defined in section 2.1.3. Values closer to one indicate straighter, more efficient travel paths. Together, these metrics provide a consistent quantitative basis for comparing alternative network configurations proposed by the optimisation algorithms.

Finally, the higher-order metric *coverage*, defined by Szell et al.[12], will also be examined. Coverage measures the proportion of the city that lies within a specified buffer distance (typically 500 m) of the bicycle network, capturing how accessible the infrastructure is to potential users. This complements purely topological indicators by evaluating the spatial reach of the proposed network and its ability to serve residential areas, workplaces, and key destinations.

Objective 3 — Agent-Based Modelling After generating candidate cycling networks using the heuristic, mathematical, and metaheuristic approaches, an agent-based model will be used to compare their real-world behavioural performance. Whereas the optimisation methods rely primarily on topological and aggregate metrics—such as connectedness, centrality, directness, and mode-specific travel times, the ABM simulates individual traveller decisions, including route choice, perceived safety, and potential shifts from car to bicycle. This enables a deeper evaluation of how each proposed network would be used in practice, providing behavioural metrics that complement the structural indicators from the earlier optimisation stages.

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